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Risk of Anterior Cruciate Ligament Tears in National Football League Players by Short, Normal, or Long Rest Weeks

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Abstract:

Introduction

Anterior cruciate ligament (ACL) tears in National Football League (NFL) players are devastating injuries that take nearly a year to recover. Players that do return to sport have worse overall performance compared to pre-ACL tear. NFL players typically play regular season games on Sunday with the next game played on the following Sunday, allowing for 7 days between games. Deviation from the usual 7-day rest week has been proposed as a potential risk for ACL tear. The main objective of this study is to evaluate the risk of decreased rest or increased rest on ACL tear rates in NFL players.

Methods

ACL injury data of NFL players from the 2012-2013 season and 2022-2023 season were gathered from publicly available sources. Player demographic data, position, age at time of injury, seasons played, injury mechanism, and playing surface type were recorded. Injuries were characterized as short, normal, or long week injuries. ACL tears that occurred during the preseason, postseason, or during week 1 were excluded. Descriptive statistics were calculated to report means, ranges, and percentages. Data were analyzed to determine statistically significant differences using Fisher's exact, chi-square, or one-way ANOVA tests.

Results

A total of 524 ACL tears were recorded in NFL players during the study window. 304 ACL tears were excluded and 220 fit inclusion criteria. 24 ACL tears occurred during short weeks, 68 during long weeks, and 128 during normal weeks. Players were 1.8 times more likely to tear their ACL during a long week compared to a normal week ($p<.001$), and 1.5 times more likely to tear their ACL during a short week compared to a normal week ($p=.02$).

Conclusion

The findings from this study suggest that deviation from the normal 7-day NFL week increases the risk of an ACL tear in NFL players when increasing or decreasing rest time. Further research exploring the impact short and long rest times on player injury risk should be conducted to prevent season ending injuries.

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Figure 1: Inclusion of ACL Tears During NFL Seasons 2012-2013 to 2022-2023

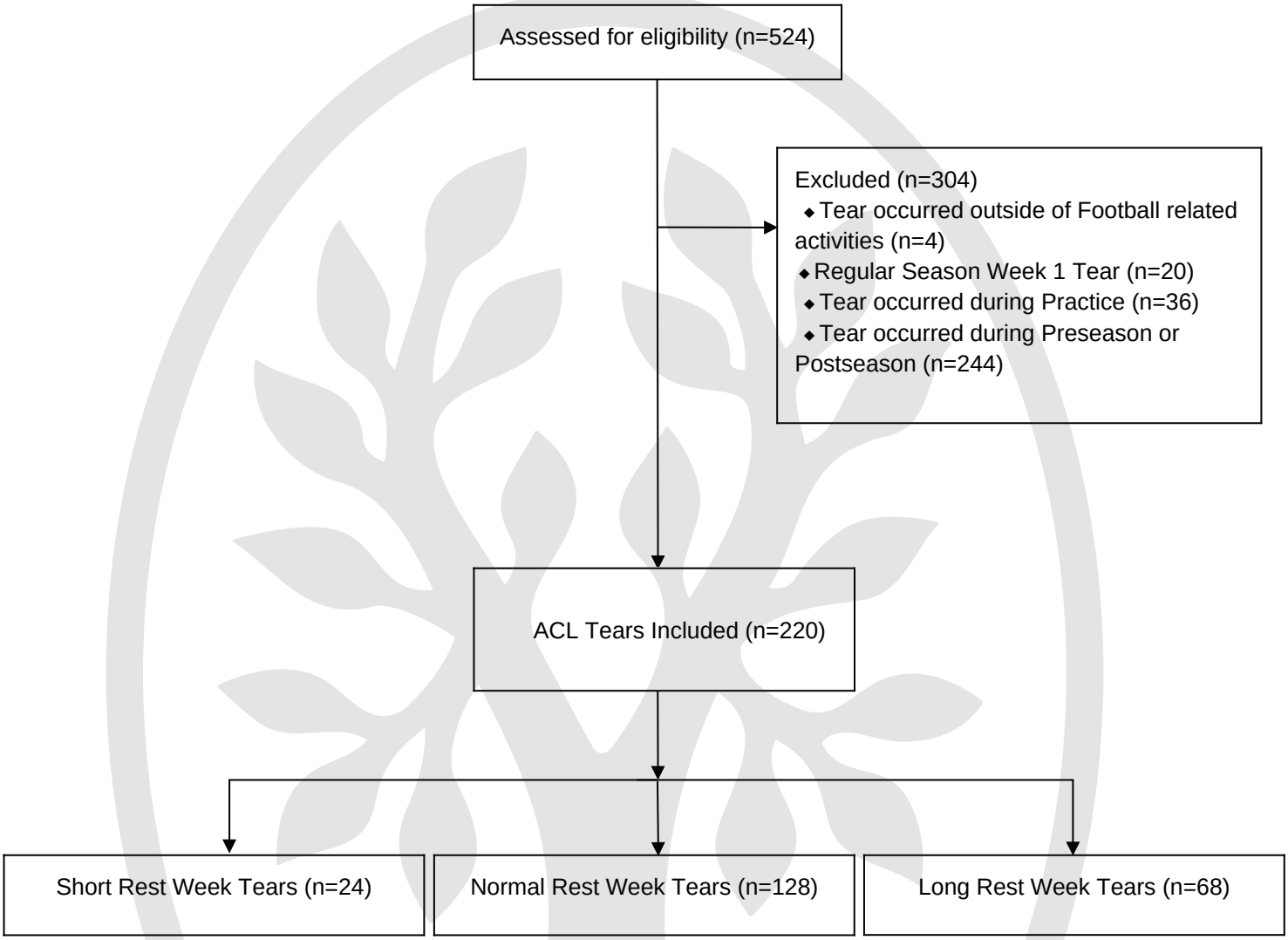

Figure Legend: NFL, National Football League. ACL, anterior cruciate ligament

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Abstract

Introduction

Anterior cruciate ligament (ACL) tears in National Football League (NFL) players are devastating injuries that take nearly a year to recover. Players that do return to sport have worse overall performance compared to pre-ACL tear. NFL players typically play regular season games on Sunday with the next game played on the following Sunday, allowing for 6 days between games. Deviation from the usual 6-day rest week has been proposed as a potential risk for ACL tear. The main objective of this study is to evaluate the risk of decreased rest or increased rest on ACL tear rates in NFL players.

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A total of 524 ACL tears were recorded in NFL players during the study window. 304 ACL tears were excluded and 220 fit inclusion criteria. 24 ACL tears occurred during short weeks, 68 during long weeks, and 128 during normal weeks. Players were 1.8 times more likely to tear their ACL during a long week compared to a normal week ($p<.001$), and 1.5 times more likely to tear their ACL during a short week compared to a normal week ($p=.02$).

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The findings from this study suggest that deviation from the normal 7-day NFL week increases the risk of an ACL tear in NFL players when increasing or decreasing rest time. Further research exploring the impact short and long rest times on player injury risk should be conducted to prevent season ending injuries.

Introduction

The anterior cruciate ligament (ACL) plays a crucial role in knee stabilization and prevents excessive motion along the anterior-posterior axis.[1] National Football League (NFL) players are at risk of ACL tear due to repetitive rapid acceleration and deceleration movements, which creates excessive strain on the ACL.[2–4] These injuries are devastating, often season-ending injuries with an average 10.8 month recovery time, with only 63% of players returning to the NFL.[5] For those that return, overall performance and production is lower compared to pre-injury, with studies showing these players start in significantly fewer games[5–7], play in fewer games, and have decreased snap counts compared to preinjury performance.[8] Of the players that came back from ACL reconstruction, only 28.5% were still playing in the NFL 3 years after their injury.[8]

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NFL players are most at risk for ACL tears if they play the wide receiver position, play on artificial turf, or play games during the first half of the NFL season.[9–16] Risk factors for injuries to NFL players are a growing focus, as rule changes can be implemented to potentially limit injury exposure and increase player safety. The NFL competition committee proposes rule changes aimed at injury prevention each year,[17,18] with previous rule changes aimed at concussion prevention and lower extremity injury prevention.[19,20] Time between games has been suggested as another potential risk factor for injury, with a few studies examining a possible correlation. Findings have been mixed, with one study finding no association, and another suggesting patients are at a lower injury risk during short weeks (<7 days between games).[21,22] Because these studies looked only at ACL tears that occurred during 2-4 seasons, additional research is needed to definitively determine if there is a correlation between number of rest days between games and ACL tears in NFL players.

To the authors knowledge, there has been no studies analyzing the risk of ACL tears based on short (less than 7 days), normal (7 days), or long (more than 7 days) rest during an NFL season. Therefore, this study aims to determine the risk of ACL tears based on short, normal, or long game weeks. We hypothesized that short and long NFL game weeks will increase the risk of ACL tear compared to normal game weeks.

Methods

Data Collection

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An online search was conducted to gather a list of NFL players who tore their ACL between the 2012-2013 season and 2022-2023 season using publicly accessible data. The 2012-2013 NFL season was selected due to an NFL lockout prior to the start of the 2011-2012 season, potentially affecting injury data. Publicly accessible resources including Pro-Football-Reference.com, ESPN.com, NFL game films from 2012-2013 to 2022-2023, and NFL team injury reports were used to accumulate the dataset.[23–25] Searches each year for "knee injury," "ACL," and "anterior cruciate ligament" were utilized to procure the injury data from the online websites and team injury reports. Player demographic data, position, age at time of injury, and seasons played was collected from Pro-Football-Reference.com. NFL injury reports, NFL.com, and other publicly available sources were used to identify which week in each season an ACL tear occurred. Mechanism of injury was determined by watching NFL game film or from a news report explicitly stating if the injury occurred due to a contact or non-contact injury mechanism. A contact injury was defined as any contact occurring immediately prior to or at the time of ACL tear. A non-contact injury was defined as an injury with no contact immediately prior to the ACL tear. ACL tears were documented to occur at practice or a game, and timing in season was recorded. Only football related injuries were included, and all ACL tears away from football related activities were excluded. Playing surface type was determined as grass or artificial turf from NFL.com and official NFL team websites at the time of injury. The playing surface was identified and recorded for each NFL game from 2012-2013 season to 2022-2023 season. Injuries were characterized as short, normal, or long week. A short week injury is an injury that occurred in which a player played in an NFL game less than 7 days after their previous NFL game. A normal week injury is an injury that occurred in which an NFL played in a game exactly

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7 days prior to their ACL tear. A long week injury is an injury that occurred in a played that played in an NFL game more than 7 days after their previous NFL game. Preseason and postseason games were not analyzed based on an inconsistency in the number of players in both settings and the number of games. Injuries occurring during week 1 were excluded due to inconsistent rest periods prior to week 1.

Statistical Analysis

Descriptive statistics were calculated to report means, ranges, and percentages. Position type grouping for analysis was as follows (defensive back, linebacker, offensive linemen, defensive linemen, quarterback, receiver, special teams, and tight ends, running backs). ACL tear risk during short, normal, and longer weeks was examined based on position group, week length, age, experience, contact vs. noncontact mechanism of injury, BMI, turf type, and whether the injury occurred before or after week 9. Fisher's exact or chi square analysis was used to examine relationship for categorical variables, and one-way ANOVA was used to examine relationships between continuous variables. Significance was set *a prior* at $p<.05$.

Results

In total, 524 ACL tears were recorded in NFL players between 2012-2013 to 2022-2023. Four ACL tears occurred outside of football related activities, 20 ACL tears occurred during week one, 36 ACL tears occurred during practice in the regular season, and 244 ACL tears occurred during the preseason or postseason. These injuries were excluded. (Figure 1) During regular season play, 220 NFL players tore their ACL between the 2012-2013 and 2022-2023 NFL seasons. With

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regards to timing, 24 ACL tears occurred during short weeks, 68 during long weeks, and 128 during normal weeks. During this same period, 5,694 games were played excluding preseason, week 1, and postseason games, with 788 being short weeks, 1429 being long weeks, and 3477 being normal weeks. This was associated with ACL tear rates of 0.037 during normal weeks, 0.030 during short weeks, and 0.048 during long weeks. Players were 1.8 times more likely to tear their ACL during a long week compared to a normal week ($p<.001$), and 1.5 times more likely to tear their ACL during a short week compared to a normal week ($p=.02$)

During the 2012-2013 to 2022-2023 NFL seasons, 220 players tore their ACL, consisting of 46 defensive backs, 33 linebackers, 66 offensive or defensive linemen, 11 quarterbacks, 29 receivers, 1 kickers, punters, or long snappers, 22 running backs, and 12 tight ends tore their ACL. Player age and body mass index with standard deviations are reported in table 1. (Table 1)

For players who suffered an ACL tear, positional grouping ($p=0.70$), offense vs. defensive grouping ($p=.806$), age ($p=.713$), contact vs. noncontact injuries ($p=.55$), BMI ($p=.64$), playing surface type ($p=.39$) and player experience ($p=.863$) were not significantly associated with an increased risk for ACL tear during short, normal, or long week. (Table 2)

Discussion

The results from this study suggest that players are 1.8 times more likely to tear their ACL during a long week compared to a normal week and 1.5 times more likely to tear their ACL during a short week compared to a normal week. No other factors, including games occurring before or after week 9, player demographics, playing surface type, or years of experience were

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significantly associated with a higher risk of ACL tear during the NFL season. These findings highlight the need for the NFL Players Association and the NFL Competition Committee to consider the risk versus benefit of short and long breaks between games, as these may place players at an increased risk for ACL tear. While short weeks in the NFL season are used to increase the number of teams that play primetime games, allowing the NFL to maximize advertising revenue, the risk to the players' health may not be worth the increased revenue.[26]

Previous studies have reported conflicting results on whether the duration of rest between games has an impact on the risk of ACL tears.[21,22,27,28] Lawrence et al. used different determinates for short, normal, and long weeks, (4, 6-8, and 10-15 days respectively) and categorized injuries into knee, ankle, hamstring, shoulder, and concussion injuries for the 2012-2013 through 2013-2014 seasons. Their study reported that across all injury types collected, there was no significant change in risk depending on the rest time between games.[21] Perez et al. analyzed injuries that resulted in a loss of playing time during NFL seasons between 2013 and 2016 and found a decreased incidence during long and short weeks compared to normal weeks.[22] This study also reports a decreased injury risk specifically for linebackers, fullbacks, and tight ends on a short week, which was not supported in the current study. Sharma et al. found that long weeks had a protective effect and led to decreases in all injury types.[28] Based on our results, a long week of rest does not have an increased risk of ACL tear compared to a normal week and may instead be associated with an increased risk of injury. The disparity of findings may be attributed to differences in injury type studied and number of seasons included in the analysis.

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Previous studies have investigated the impact of fatigue on ACL tear risk and report that fatigue is significantly associated with an increased incidence of ACL tears.[29] Many NFL players have been outspoken about a dislike for shortened rest weeks.[30] During a normal week, days 5 and 6 allow for decreased practice intensity and improved recovery before gameday. Alternatively, short, or long weeks impede the predictable schedule, potentially depriving players of rest and recovery days, or throwing off recovery schedules such that player readiness before games is impacted. Soreness and fatigue are difficult variables to account for but could affect the body mechanics during running and cutting, ultimately increasing the risk of ACL tear. Short weeks in the NFL season may also be used as a mini-bye week. If an NFL game is played on a Thursday and then the next game is played on the next Sunday, this creates 9-days of time between games. A short week must be followed by a long week of rest in this case. This extra time away from football activities seems to be preferred by players.[31]

Increased rest past the normal 7-day NFL playing week may lead to deconditioning. Following the NFL lockout in 2011, Achilles tendon ruptures were four times higher compared to prior seasons.[32] The 2011 NFL lockout prohibited any organized team activities, leading to a four-month break from football. Although this deconditioning period was longer, there was a significant increase in Achilles tendon ruptures. Extended rest and time away from football during a >7-day NFL playing week may contribute to deconditioning and contribute to an increased risk of ACL tear.

A limitation of this study is that the data was collected from publicly accessible databases such as NFL injury reports, so it is possible that all the information is not present. NFL teams are

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required to publish weekly injury reports, listing all minor to major injuries. An ACL tear is usually a season ending injury, so it is expected that the number of unreported injuries are low. Also, the length of rest in days may not be representative of the rest a player truly receives. Depending on the team and player, the level of strenuous activity between games could be different and these stresses could play a major role in frequency of injuries. These limitations are not expected to affect the primary outcome of this study.

Conclusion

From the 2012-2013 to 2022-2023 NFL seasons, 220 ACL tears occurred over 5,694 games. NFL players experienced significantly higher rates of ACL tears during short (<7 days) and long (>7 days) weeks of rest between games. NFL player position, age, experience, mechanism of injury, or in season injury timing was not found to be significantly associated with an increased risk for injury based on short, normal, or long weeks. The NFL should continue to explore the impact short and long rest times have on player injury risk to determine if changes should be implemented to prevent these season ending injuries.

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Table 1: Player Position, Age, and BMI

Position	Number of ACL Tears	Age (sd)	BMI (sd)
Defensive Backs	46	24.9 (3.09)	27.0 (1.27)
Linebackers	33	25.0 (2.47)	31.8 (2.16)
Offensive Linemen	38	26.3 (3.33)	36.4 (1.96)
Defensive Linemen	28	25.2 (3.45)	31.9 (3.73)
Quarterbacks	11	26.4 (3.04)	27.5 (1.32)
Receivers	29	24.7 (3.05)	26.2 (1.35)
Running Backs	22	24.0 (3.57)	29.8 (1.49)
Tight Ends	12	25.1 (2.43)	29.5 (1.02)
Specialists	1	22	27.5

Key: ACL-anterior cruciate ligament, sd – standard deviation, BMI- body mass index

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Table 2: Age, BMI, Experience, Offense vs. Defense, Playing Surface, and Mechanism of Injury Characterized by Short, Normal, and Long Weeks

	Short Week (<7 days between NFL games)	Normal Week (7 days between NFL games)	Long Week (>7 days between NFL games)	p Value
Age*	25.14	25.21	25.16	0.713
BMI*	30.74	30.8	30.77	0.64
Experience*	3.29	3.36	3.29	0.863
Offense vs. Defense				
Offense	26	62	24	0.806
Defense	18	64	25	
Playing Surface				
Grass	21	59	27	0.39
Turf	23	68	22	
Mechanism of Injury				
Contact	30	84	29	0.55
Noncontact	10	35	16	

Key: (*)-reported as averages, BMI-body mass index

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